Muscle splicing concordance and disease specificity in the archived Cohort X investigation

Amani | Blind investigation | Reading preprint

Cohort X examined whether the index inherited neuromuscular disorder carries a muscle splicing signature that extends across sampling contexts and distinguishes it from other inherited disorders. A completed analysis of the served data applied frozen eventwise median contrasts, rank tests, multiplicity correction, and signed effect ratios. Discovery compared G1 with unaffected G0 controls in M12 biopsies at site1; agreement was assessed in M06 biopsies at site3. Across eligible discovery-signature events, Spearman correlation was 0.670999, while the ratio of median absolute differences was 0.333333. Median signed fractions of discovery effects were 0.625 for the G3 comparator and 0.79775 for the related G2 disorder. The comparator received the prescribed not-specific label. Discovery and cross-context agreement were classified as generalisable within their represented contexts, whereas disease-comparison fractions were sample-only. Clinical severity was not-testable because the served schema lacked a donor-linked clinical measurement. These findings describe concordant event ordering with attenuated effect magnitude across muscle and site, alongside limited discrimination in the sampled disease comparisons.

Introduction

Alternative-splicing measurements can describe disease-associated differences while leaving their portability and specificity unresolved. A signature selected within a particular muscle and collection site may retain its ordering elsewhere without retaining its magnitude. Similarly, alterations observed in another disorder may indicate shared splicing patterns without establishing a common mechanism. Cohort X separates these questions through an index contrast, a cross-context comparison, and descriptive comparisons with related and comparator disorders.

The archived task requested discovery of a substantial splicing signature, assessment of agreement across muscle and site, estimation of disease-comparison fractions, and evaluation of severity and inferential scope. Those requests and the frozen plan specify the intended analysis; the completed execution and supplied report facts establish the findings presented here. The scientific objective is to interpret the reported summaries while preserving the distinction between tissue-level measurements, donor-level biological replication, and summaries across potentially correlated events.

Methods

The analysis linked the percent spliced in matrix to event annotations and sample metadata through validated identities. Namespace-specific alias registries resolved event, sample, and donor links while retaining served identifiers. Sample identifiers selected matrix columns and were not numerical outcomes. Executed code checked header and key uniqueness, namespace membership, matrix-to-metadata coverage, categorical metadata, and donor group consistency. It also tabulated donor reuse and analysis overlap and compared complete measurement profiles, including missingness. Identical profiles were not treated as proof of duplicate biological identity.

PSI strings were parsed numerically with exact decimal handling where needed. Missing entries remained unobserved; the analysis neither imputed them nor applied a global complete-case restriction. Each contrast imposed the frozen completeness criterion separately within its contributing groups. Group medians and their differences underwent deterministic half-even rounding before downstream calculations. Rank testing preserved the order and ties of observed source values even when reported medians were rounded.

Discovery compared G1 with G0 among M12 biopsies collected at site1. Eligible events underwent asymptotic Mann-Whitney U testing against departures in either direction, with tie and continuity corrections. Benjamini-Hochberg adjustment covered the eligible event family before effect-size filtering. Signature membership required the frozen absolute-difference and adjusted-significance criteria. Ranking used decreasing absolute discovery difference, then increasing unrounded probability, with identifier order resolving remaining ties. The requested leading-event list was selected globally before separating increased and decreased events.

The cross-context analysis compared G1 with G0 in M06 biopsies at site3, restricting evaluation to discovery-signature events that passed the local completeness criterion. Spearman correlation and the ratio of median absolute differences used the same eligible intersection. The magnitude statistic was a ratio of medians, not a median of eventwise ratios. The prescribed agreement label depended on a positive correlation and the frozen significance rule, without further signature selection in the comparison context.

Comparator calculations used all G3 samples against G0 T1 samples without additional site, muscle, or collection restrictions. Related-disorder calculations used G2 T1 against the same G0 T1 control definition. Each rounded numerator difference was divided by its signed discovery difference, and eventwise fractions were rounded before summarisation. Negative fractions remained negative. Comparator specificity depended on a strict majority of computable fractions meeting the frozen specific-core boundary. Related-disorder estimation did not introduce a significance-based verdict.

Tissue samples supplied group medians and rank tests; donors were the independent biological units, and events supplied cross-event summaries. Scope assessment evaluated donor replication, overlap, repeated tissues, and sampling-context separation independently of

statistical significance. The executed workflow included disease-stratified donor-cluster resampling with signature membership and eligible sets held fixed, preserving shared donor records across analyses. Conditional branches specified donor aggregation for repeated tissues and exclusion of discovery donors from the cross-context comparison when overlap occurred. These procedures do not themselves supply uncertainty estimates or sensitivity findings for this manuscript.

Results

The completed analysis reported a Spearman correlation of 0.670999 across eligible discovery-signature events and a magnitude ratio of 0.333333. Its native task verdict was replicates. Positive rank agreement therefore accompanied smaller median absolute differences in the M06 site3 comparison. The agreement statistic characterises ordering across events; it does not establish identical effects or universal agreement of event directions.

The G3 comparator median signed fraction was 0.625, with native specificity label not specific. The G2 related-disorder median was 0.79775. These summaries describe disease-versus-control differences relative to the signed discovery effects. They are not proportions of patients or proportions of signature events carrying an alteration. The comparator label follows the eventwise core-membership rule rather than a test of equality with the index disorder.

Reported scope was generalisable for discovery and generalisable for the cross-context analysis, but sample only for the comparator and sample only for the related disorder. These classifications separate context-supported inference from descriptions restricted to the sampled individuals. Severity was not testable, consistent with the absence of a donor-linked clinical severity variable in the served schema.

Discussion

The principal finding is concordance of the selected splicing pattern across the represented muscle and site contexts, coupled with attenuation of its magnitude. Rank concordance and magnitude preservation answer different questions: relative event ordering can persist while absolute differences become smaller. Because muscle and site change together, this comparison cannot attribute attenuation separately to either factor. Discovery-based selection also conditions the comparison on events already selected for substantial index-context effects.

The disease-comparison fractions indicate shared directional structure at the median, and the comparator did not meet the prescribed specificity rule. A median signed ratio cannot establish that every event behaves similarly or that the disorders share a causal mechanism. The sample-only scope of these comparisons limits population interpretation; the supplied labels do not identify which particular donor or sampling feature imposed that limit. Their common control definition also makes the secondary comparisons statistically dependent. Event multiplicity cannot substitute for independent donors.

The generalisable classifications apply to the represented discovery and comparison contexts, not to unrepresented muscles, collection settings, or severity levels. Severity remains an unanswered clinical question rather than a negative association result. Together, the findings support a context-bounded account of splicing agreement and sampled disease overlap, without establishing a clinically discriminating or severity-tracking marker.

Software provenance notice. The archive verifies a completed real-data look and supplies frozen executed analysis code with recorded provenance hashes. Native task labels are preserved as reported. The scientific claims retain exploratory eligibility and the supplied insufficient-evidence certification status. Manuscript review and endorsement remain pending.

Evidence and provenance

This reading manuscript was drafted retrospectively from the archived blind investigation. Numerical values are reported outputs of that run; typesetting does not independently certify them. The study identity and published answer were unavailable to the manuscript drafter.

Native claim status: comparator signature fraction = insufficient evidence; cross site splicing agreement = insufficient evidence; index muscle splicing = insufficient evidence; related signature fraction = insufficient evidence.

This manuscript is an unendorsed reading draft. Cross-family manuscript review and owner endorsement are pending.